



Investigating Novel Predictions of Hypoglycemia Occurrence using Rreal-world Models

Dissemination Portfolio

13 May 2025

iNPHORM Principal Investigators

Alexandria Ratzki-Leewing, PhD, MSc

Assistant Professor

Dept. of Epidemiology and Public Health

University of Maryland (UM) School of Medicine

UM Institute for Health Computing

Stewart Harris, CM, MD, MPH, FCFP, FACPM

Professor Emeritus

Dept. of Family Medicine

Schulich School of Medicine & Dentistry

Western University

Executive Summary

iNPHORM is the **first prospective, population-based study of hypoglycemia** incidence, risk factors, and outcomes in US adults (≥ 18 years old) with type 1 diabetes or insulin/secretagogue-treated type 2 diabetes. The study **leverages the value of self-reported data** to expose the true real-world burden of hypoglycemia. Prior population research relying on health service records has been shown to capture $<10\%$ of all Level 3 hypoglycemia and virtually no non-severe events. **Over 1,200 participants were recruited from a probability-based internet panel** designed to reflect the geodemographic, attitudinal, and behavioral distribution of the US public. A screener, baseline, and up to 12 monthly follow-up **questionnaires collected data on >600 unique variables** between 2020 to 2021. *Findings have been cited in the 2023 and 2025 American Diabetes Association (ADA) Standards of Care in Diabetes, informing today's strategies to reduce treatment-related harm.*

Key Indicators

Output

- ⇒ **7** Published Manuscripts
- ⇒ **25** Published Abstracts (*4 Encore*)
- ⇒ **32** Selected Conference Presentations (*4 Encore*)
- ⇒ **15** Invited Symposia and Major Lectures (*1 Encore*)
- ⇒ **12** Media Features

Impact

- ⇒ **1** article cited in the ADA Standards of Care in Diabetes, “**4. Comprehensive Medical Evaluation and Assessment of Comorbidities**”—**2023**
- ⇒ **1** article cited in the ADA Standards of Care in Diabetes, “**6. Glycemic Goals and Hypoglycemia**”—**2025**

Table of Contents

<i>Grant Support.....</i>	<i>4</i>
<i>Published Peer-Reviewed Manuscripts.....</i>	<i>5</i>
<i>Published Theses.....</i>	<i>6</i>
<i>Selected Presentations</i>	<i>6</i>
Published proceedings.....	6
Unpublished proceedings.....	9
<i>Invited Symposia and Major Lectures.....</i>	<i>10</i>
<i>Media Features.....</i>	<i>12</i>

Grant Support

<i>Title:</i>	Investigating Novel Predictions of Hypoglycemia Occurrence Using Real-World Models (iNPHORM), US
<i>PIs:</i>	Ratzki-Leewing A and Harris S
<i>Co-Is:</i>	Ryan B, Zou GY
<i>Funder:</i>	Sanofi Global
<i>Grant type:</i>	Investigator Sponsored Study
<i>Funding:</i>	\$1.3 million (CAD)
<i>Funding period:</i>	Sept 2019 – Sept 2021
<i>Summary:</i>	A 1-year prospective survey study on real-world hypoglycemia risk among American adults with diabetes taking insulin and/or secretagogues.
<i>Title:</i>	Unpacking the Intersection of COVID-19 & Diabetes (iNPHORM), US
<i>PIs:</i>	Ratzki-Leewing A and Harris S
<i>Co-Is:</i>	Ryan B, Zou GY
<i>Funder:</i>	Sanofi Global
<i>Grant type:</i>	Investigator Sponsored Study
<i>Funding:</i>	\$49,816 (CAD)
<i>Funding period:</i>	Jan 2021 – Nov 2021
<i>Summary:</i>	A US investigation of the impacts of COVID-19 on diabetes management.
<i>Title:</i>	Investigating Novel Predictions of Hypoglycemia Occurrence Using Real-World Models (iNPHORM): Research Outgrowth, US
<i>PIs:</i>	Ratzki-Leewing A and Harris S
<i>Co-Is:</i>	Ryan B, Zou GY
<i>Funder:</i>	Sanofi Global
<i>Grant type:</i>	Investigator Sponsored Study
<i>Funding:</i>	\$96,970.00 (CAD)
<i>Funding period:</i>	Jun 2022 – Jun 2023
<i>Summary:</i>	An extension of iNPHORM that includes the production and publication of 5 manuscripts, as well as related knowledge exchange.

Published Peer-Reviewed Manuscripts

1. Ratzki-Leewing A, Ryan B, Buchenberger J, Dickens J, Black JE, Harris S. COVID-19 hinterland: Surveilling the self-reported impacts of the pandemic on diabetes management in the USA (cross-sectional results of the iNPHORM study). *BMJ Open*, 2021; 11(9):e049782.
 - ***Referenced in ADA Standards of Care in Diabetes, “4. Comprehensive Medical Evaluation and Assessment of Comorbidities”—2023***
2. Ratzki-Leewing A, Ryan BL, Zou GY, Webster-Bogaert S, Black JE, Stirling K, Timcevska K, Khan N, Buchenberger JD, Dickens JW, Harris S. Predicting real-world hypoglycemia risk in American adults with type 1 or 2 diabetes mellitus prescribed insulin and/or secretagogues: Protocol for a prospective, 12-wave internet-based panel survey with email support (the iNPHORM [Investigating Novel Predictions of Hypoglycemia Occurrence Using Real-world Models] Study). *J Med Internet Res*, 2022; 11(2):e33726.
3. Ratzki-Leewing A, Black JE, Ryan BL, Harris S. Real-world risk factors of confirmed or probable COVID-19 infection in Americans with diabetes: A prospective, community-based study (iNPHORM). *Endocrinol Diabetes Metab*, 2022; 5(4):e342.
4. Black JE, Harris SB, Ryan BL, Zou GY, Ratzki-Leewing A. Real-world effects of second-generation versus earlier intermediate/basal insulin analogues on rates of hypoglycemia in adults with type 1 and 2 diabetes (iNPHORM, US). *Diabetes Ther*. 2023; 4(8):1299-1317. doi: 10.1007/s13300-023-01423-3.
5. Ratzki-Leewing A, Black JE, Ryan BL, Zou GY, Klar N, Webster-Bogaert S, Timcevska K, Harris S. Development and validation of a real-world model to predict one-year, recurrent Level 3 (severe) hypoglycaemia risk in adults with diabetes (the iNPHORM study, United States). *Diabetes Obes Metab*, 2023; 25(10):2910-2927.
6. Ratzki-Leewing A, Black JE, Kahkoska AR, Ryan BL, Zou GY, Klar N, Timcevska K, Harris SB. Level 3 (severe) hypoglycaemia in a real-world cohort of adults with type 1 or 2 diabetes mellitus (iNPHORM, United States). *Diabetes Obes Metab*, 2023; 25(12):3736-3747.
 - ***Referenced in ADA Standards of Care in Diabetes, “6. Glycemic Goals and Hypoglycemia”—2025***
7. Lin YK, Black JE, Harris SB, Ryan BL, Zou GY, Ratzki-Leewing A. Young adults with type 2 diabetes experience high rates of level 3 hypoglycemia: A subgroup analysis of the real-world iNPHORM cohort. *Diabetes Res Clin Pract*, 2025; *In press*.

Published Theses

PhD Dissertation: Ratzki-Leewing A. Quantifying and predicting real-world iatrogenic severe hypoglycemia in adults with type 1 or 2 diabetes mellitus (the iNPHORM study, United States). 2022; *Electronic Thesis and Dissertation Repository*. 9064. <https://ir.lib.uwo.ca/etd/9064>

Selected Presentations

Published proceedings

1. Ratzki-Leewing A, Ryan BL, Zou GY, Au NH, Webster-Bogaert S, Harris S. Investigating novel predictions of hypoglycemia occurrence using real-world models (iNPHORM Study): A study protocol. Morressier, 2019.
 - **Poster Presentation** at *The International Diabetes Federation Congress*. Busan, KOR. Dec 2nd-6th, 2019.
2. Ratzki-Leewing A, Harris SB, Zou G, Ryan BL. Real-world estimates of severe hypoglycaemia and associated healthcare utilisation in the US: Baseline results of the iNPHORM study. *Diabetologia*, 2020; 63(Suppl 1):750P, S363.
 - **Moderated Poster Presentation** at *The 56th Annual Meeting of the European Association for the Study of Diabetes*. Virtual Meeting. Sept 22nd-25th, 2020.
3. **ENCORE** – Ratzki-Leewing A, Harris S, Ryan B, Zou GY. Real-world estimates of severe hypoglycemia and associated healthcare utilization in the US: Baseline results of the iNPHORM study. *Diabetes Technol & Ther*, 2021; 23(Supplement 2):A-140.
 - **Poster Presentation** at *The 14th International Conference on Advanced Technologies & Treatments for Diabetes*. Virtual Meeting. Jun 2nd-25^h, 2021.
4. Ratzki-Leewing A, Harris S, Black J, Stirling K, Zou G, Ryan B. Second-generation basal insulin analogues: First choice in reducing real-world severe hypoglycemia (baseline results of the iNPHORM study, USA). *Diabetes Technol & Ther*, 2021; 23(Supplement 2):A-89 to A-90.
 - **Poster Presentation** at *The 14th International Conference on Advanced Technologies & Treatments for Diabetes*. Virtual Meeting. Jun 2nd-25^h, 2021.
5. Ratzki-Leewing A, Black JE, Ryan BL, Zou GY, Harris SB. Why some Americans use healthcare following severe hypoglycemia, and why some do not: Baseline results of the iNPHORM study. *Diabetes*, 2021; 70(Supplement 1):347-P.
 - **Poster Presentation** at *The American Diabetes Association's 81st Scientific Sessions*. Virtual Meeting. Jun 25th-29th, 2021.

6. Ratzki-Leewing A, Harris S, Black JE, Ryan BL, Zou GY. How COVID-19 has impacted diabetes management in the United States (iNPHORM study). *Diabetologia*, 2021; 64(Suppl.1):S362.
 - **Oral Presentation** at *The 57th Annual Meeting of the European Association for the Study of Diabetes*. Virtual Meeting. Sept 27th-Oct 1st, 2021.
7. **ENCORE** – Ratzki-Leewing A, Harris S, Black JE, Khan N, Timcevska K, Ryan BL, Zou GY. COVID-19 impact on diabetes management (iNPHORM, USA). *Can J Diabetes*, 2021; 45(7):S3.
 - **Oral Presentation** at *The 24th Diabetes Canada/Canadian Society of Endocrinology and Metabolism Professional Conference*. Virtual Meeting. Nov 23rd-26th, 2021.
8. Ratzki-Leewing A, Harris SB, Webster-Bogaert S, Black JE, Zou GY, Ryan BL. Real-world severe hypoglycemia incidence in Canada and the United States. *Diabetes Res Clin Pract*, 2021; 186(Suppl 1):109739.
 - **Oral Presentation** at *The International Diabetes Federation Congress 2021. Virtual Meeting*. December 6th-11th, 2021
9. Ratzki-Leewing A, Harris S, Zou GY, Ryan B. A tool not a treatment: the effect of long-term continuous/flash glucose monitoring on real-world hypoglycemia rates (iNPHORM study). *Diabetes Technol Ther*, 2022; 24(Suppl 1):A-34.
 - **Oral Presentation** at *The 15th International Conference on Advanced Technologies & Treatments for Diabetes*. Virtual Meeting. Apr 27th-30th, 2022.
10. Ratzki-Leewing A, Harris SB, Black JE, Zou GY, Webster-Bogaert S, Ryan BL. Predicting real-world severe hypoglycemia in Americans with diabetes (iNPHORM). *Diabetes*, 2022; 71(Suppl 1):372–P.
 - **Poster Presentation** at *The American Diabetes Association's 82nd Scientific Sessions*. New Orleans, LA. Jun 3rd-7th, 2022.
11. Ratzki-Leewing A, Harris SB, Black JE, Zou GY, Webster-Bogaert S, Ryan BL. Predicting real-world non-severe hypoglycemia in Americans with diabetes (iNPHORM). *Diabetes*, 2022; 71(Suppl 1):371–P.
 - **Poster Presentation** at *The American Diabetes Association's 82nd Scientific Sessions*. New Orleans, LA. Jun 3rd-7th, 2022.
12. Ratzki-Leewing A, Black J, Kahkoska A, Ryan BL, Harris SB. Real-world estimates of severe hypoglycemia and associated healthcare utilization in the US: Prospective, longitudinal results of the iNPHORM study. *Diabetologia*, 2022; 65(Suppl 1):S323.
 - **Short Oral Presentation** at *The 58th Annual Meeting of the European Association for the Study of Diabetes*. Stockholm, SE. Sept 19th-23rd, 2022.

13. **ENCORE** – Ratzki-Leewing A, Harris SB, Black JE, Zou GY, Webster-Bogaert S, Timcevska K, Ryan BL. Predicting real-world non-severe hypoglycemia risk in diabetes (iNPHORM, USA). *Can J Diabetes*, 2022; 46(7)(Suppl 28).
 - **Poster Presentation** at *The 25th Diabetes Canada/ Canadian Society of Endocrinology and Metabolism Professional Conference*. Calgary, AB. Nov 12th, 2022.
14. **ENCORE** – Ratzki-Leewing A, Harris SB, Black JE, Zou GY, Webster-Bogaert S, Timcevska K, Ryan BL. Predicting real-world severe hypoglycemia risk in diabetes (iNPHORM, USA). *Can J Diabetes*, 2022; 46(7)(Suppl 28).
 - **Oral Presentation** at *The 25th Diabetes Canada/Canadian Society of Endocrinology and Metabolism Professional Conference*. Calgary, AB. Nov 12th, 2022.
15. Ratzki-Leewing A, Black J, Zou G, Ryan B, Harris S. Continuous/flash glucose monitoring does not prevent severe hypoglycemia in real-world settings (iNPHORM, USA). *Diabetes Technol Ther*, 2023; 25(S2):A-64.
 - **Oral Presentation** at *The 16th International Conference on Advanced Technologies & Treatments for Diabetes*. Berlin, DE. Feb 22nd-25^h, 2023.
16. Ratzki-Leewing A, Black JE, Zou GY, Ryan BL, Harris SB. High incidences of Level 3 severe hypoglycemia reported by T2DM secretagogue-users (iNPHORM, USA). *Diabetes*, 2023; 72(Supplement_1):387-P.
 - **Poster Presentation** at *The American Diabetes Association's 83rd Scientific Sessions*. San Diego, CA. Jun 23rd-26th, 2023.
17. Ratzki-Leewing A, Black JE, Zou GY, Ryan BL, Harris SB. Severe hypoglycemia among older adults with diabetes (iNPHORM, USA). *Diabetes*, 2023; 72(Supplement_1):3-OR.
 - **Oral Presentation** at *The American Diabetes Association's 83rd Scientific Sessions*. San Diego, CA. Jun 23rd-26th, 2023.
18. Ratzki-Leewing A, Black JE, Ryan BL, Zou GY, Harris SB. Food insecurity doubles the rate of severe hypoglycemia in adults with diabetes: Real-world results from the iNPHORM Study (United States). *Diabetologia*, 2023; 66(Suppl 1):750.
 - **Short Oral Presentation** at *The 59th Annual Meeting of the European Association for the Study of Diabetes*. Hamburg, DE. Oct 2nd-6th, 2023.
19. Harris SB, Black JE, Ryan BL, Zou GY, Ratzki-Leewing A. Hypoglycemia self-management among adults with diabetes: Gaps and leverage points (iNPHORM study, United States). *Diabetologia*, 2023; 66(Suppl 1):744.
 - **Short Oral Presentation** at *The 59th Annual Meeting of the European Association for the Study of Diabetes*. Hamburg, DE. Oct 2nd-6th, 2023.

20. Ratzki-Leewing A, Black JE, Saeed H, Zou GY, Ryan BL, Harris SB. Differential effect of insulin pump use on real-world level 3 hypoglycemia rates in adults with type 1 or 2 diabetes mellitus (iNPHORM, United States). *Diabetes Technol Ther*, 2024; 26(S2): AS07.
 - **Oral Poster Presentation** at *The 17th International Conference on Advanced Technologies & Treatments for Diabetes*. Florence, ITALY. Mar 6th-9th, 2024.
21. Ratzki-Leewing A, Black JE, Saeed H, Zou GY, Ryan BL, Harris SB. Better safe than sorry: Should older adults use CGM? *Diabetes Technol Ther*, 2024; 26(S2): O0850.
 - **Oral Presentation** at *The 17th International Conference on Advanced Technologies & Treatments for Diabetes*. Florence, ITALY. Mar 6th-9th, 2024.
22. Ratzki-Leewing A, Black JE, Zou GY, Ryan BL, Harris SB. How common is fear of hypoglycemia in T1DM and T2DM? (iNPHORM, USA). *Diabetes*, 2024; 73(Supplement_1):369-P.
 - **Poster Presentation** at *The American Diabetes Association's 84th Scientific Sessions*. Orlando, FL. Jun 21st-24th, 2024.
23. Ratzki-Leewing A, Black JE, Zou GY, Ryan BL, Harris SB. Level 3 hypoglycemia in diabetes: An age-old but not (just) old-age problem (iNPHORM, USA). *Diabetes*, 2024; 73(Supplement_1):175-OR.
 - **Oral Presentation** at *The American Diabetes Association's 84th Scientific Sessions*. Orlando, FL. Jun 21st-24th, 2024.
24. Ratzki-Leewing A, Black J, Ryan BL, Zou GY, Harris SB. Real-world prevalence and risk factors of diabetes ketoacidosis in adults with type 1 diabetes (iNPHORM, USA). *Diabetes Technol Ther*, 2025; 27(S2): AS21.
 - **Short Oral Poster Presentation** at *The 18th International Conference on Advanced Technologies & Treatments for Diabetes*. Amsterdam, NL. Mar 19th-22nd, 2025.
25. Ratzki-Leewing A, Black JE, Kahkoska A, Gandhi K, Ryan BL, Zou GY, Harris SB. Sex, age, and diabetes type: Dissecting patterns of real-world level 3 hypoglycemia (iNPHORM, USA). *Diabetes*, 2025; *In press*.
 - **Poster Presentation** at *The American Diabetes Association's 85th Scientific Sessions*. Chicago, IL. Jun 20th-23rd, 2025.

Unpublished proceedings

1. Ratzki-Leewing A, Harris S. Real-World Deficiencies in Hypoglycemia Self-Management Behaviour by Diabetes Type: Results from the iNPHORM Study. **Oral Presentation** at *The 1st Annual Advances in Hypoglycaemia Scientific Conference*, International Hypoglycaemia Study Group, Virtual Meeting, 2021.
 - **Awarded Best Oral Presentation**

2. Bhatti J, Black JE, Fernandes K, Hamdan J, Harris SB, Zou GY, Ryan BL, Ratzki-Leewing A. Level 3 hypoglycemia frequency by type of insulin secretagogue use (iNPHORM, United States). **Short Oral Presentation** at *The 3rd Annual Advances in Hypoglycaemia Scientific Conference*, International Hypoglycaemia Study Group, Virtual Meeting, 2024.
3. Hamdan J, Black JE, Bhatti J, Fernandes K, Harris SB, Zou GY, Ryan BL, Ratzki-Leewing A. Iatrogenic hypoglycemia symptoms among adults with type 1 diabetes (T1D) or type 2 diabetes (T2D) (iNPHORM, United States). **Short Oral Presentation** at *The 3rd Annual Advances in Hypoglycaemia Scientific Conference*, International Hypoglycaemia Study Group, Virtual Meeting, 2024.
4. Fernandes K, Black JE, Hamdan J, Bhatti J, Harris SB, Zou GY, Ryan BL, Ratzki-Leewing A. Hypoglycemia management attitudes and Level 3 hypoglycemia incidence in adults with diabetes (iNPHORM, United States). **Short Oral Presentation** at *The 3rd Annual Advances in Hypoglycaemia Scientific Conference*, International Hypoglycaemia Study Group, Virtual Meeting, 2024.
5. Black J, Harris SB, Ryan BL, Zou GY, Ratzki-Leewing A. Real-world effect of 2nd generation vs other basal insulin analogues on hypoglycemia risk. **Oral Presentation** at *The 3rd Annual Advances in Hypoglycaemia Scientific Conference*, International Hypoglycaemia Study Group, Virtual Meeting, 2024.
6. Fernandes KM, Black JE, Zou GY, Ryan BL, Harris SB, Ratzki-Leewing A. Examining Intentional Blood Glucose Elevation and A1C Levels Among Adults with Diabetes (iNPHORM, USA). **Short Oral Presentation** at *The 4th Annual Advances in Hypoglycaemia Scientific Conference*, International Hypoglycaemia Study Group, Virtual Meeting, 2025.
7. Chun S, Bhatti J, Black JE, Persaud F, Bokyana J, McCoy RM, Zou GY, Ryan BL, Harris SB, Ratzki-Leewing A. Rate and Risk Factors of Level 3 Hypoglycemia in Older Adults with Diabetes (iNPHORM, USA). **Short Oral Presentation** at *The 4th Annual Advances in Hypoglycaemia Scientific Conference*, International Hypoglycaemia Study Group, Virtual Meeting, 2025.

Invited Symposia and Major Lectures

1. Ratzki-Leewing A. Investigating Novel Predictions of Hypoglycemia Occurrence Using Real-World Models (iNPHORM). Industry Seminar, Sanofi Global, Virtual Meeting, 2019.
2. Ratzki-Leewing A, Harris S. Investigating Novel Predictions of Hypoglycemia Occurrence Using Real-World Models (iNPHORM): Updates. Industry Seminar, Sanofi Global, Virtual Meeting, 2020.

3. Ratzki-Leewing A, Harris S. Investigating Novel Predictions of Hypoglycemia Occurrence Using Real-World Models (iNPHORM): COVID-19. Industry Seminar, Sanofi Global, Virtual Meeting, 2020.
4. Ratzki-Leewing A, Harris S. Investigating Novel Predictions of Hypoglycemia Occurrence Using Real-World Models (iNPHORM): Updates. Industry Seminar, Sanofi Global, Virtual Meeting, 2020.
5. Ratzki-Leewing A, Harris S. Investigating Novel Predictions of Hypoglycemia Occurrence Using Real-World Models (iNPHORM): COVID-19. Industry Seminar, Sanofi Global, Virtual Meeting, 2020.
6. Ratzki-Leewing A, Harris S. Real-World Diabetes-Related Hypoglycemia in the US: Results of the iNPHORM Study. Grand Rounds, Division of Endocrinology and Metabolism, Department of Medicine, Schulich School of Medicine & Dentistry, Western University, London, ON, 2021.
7. Ratzki-Leewing A, Harris S. Investigating Novel Predictions of Hypoglycemia Occurrence Using Real-World Models (iNPHORM): Update. Industry Seminar, Sanofi-Aventis United States/Sanofi Global, Virtual Meeting, 2021.
8. Ratzki-Leewing A, Harris S. The Real-World Burden and Cost of Diabetes: Results from the iNPHORM Study. Symposium Session, The 57th Annual Meeting of the European Association for the Study of Diabetes, Virtual Meeting, 2021.
9. Ratzki-Leewing A, Harris S. Investigating Novel Predictions of Hypoglycemia Occurrence Using Real-World Models (iNPHORM): Update. Industry Seminar, Sanofi-Aventis United States/Sanofi Global, Virtual Meeting, 2022.
10. Ratzki-Leewing A. A tool not a treatment: The effect of long-term continuous/flash glucose monitoring on real-world hypoglycemia rates (iNPHORM Study). On-Demand Lecture, VuMedi, 2022.
11. Ratzki-Leewing A. From research branding to global influence: The case of the iNPHORM Study. Grand Rounds, Centre for Studies in Family Medicine, Department of Family Medicine, Schulich School of Medicine & Dentistry, Western University, London, ON, 2024.
12. **ENCORE** – Ratzki-Leewing A. From Research Branding to Global Influence: The Case of the iNPHORM Study. Pepper Center Monthly Rounds, University of Maryland School of Medicine, Baltimore, MD, 2024.
13. Ratzki-Leewing A. From real-world data to anti-fake world evidence: The story of iNPHORM. Grand Rounds, Department of Epidemiology and Biostatistics, Schulich School of Medicine & Dentistry, Western University, London, ON, 2024.

14. Ratzki-Leewing A. What is iNPHORM? Population Health Rounds, University of Maryland Institute for Health Computing, North Bethesda, MD, 2024.
15. Ratzki-Leewing A, Harris S. The Importance of Hypoglycaemia in the Real World. Symposium Session, The 60th Annual Meeting of the European Association for the Study of Diabetes. Madrid, ES. Sept 10th-13th, 2024.

Media Features

1. Western University. The real low-down on diabetes (iNPHORM Study). *Schulich School of Medicine and Dentistry e-newsletter*. <https://tinyurl.com/y892pm4u>. 2019. [Press release]
2. Western University. Unique study uncovers realities of hypoglycemia. *Western News*. <https://tinyurl.com/yb6s9vmt>. 2021. [News article]
3. Tucker, ME. 'Hidden Toll' of Pandemic a Major Deficiency in Diabetes Care. *Medscape*. <https://tinyurl.com/yc8mya2h>. 2021. [News article]
4. EurekAlert! EASD Media Office. US Survey highlights considerable gaps in diabetes care during COVID-19 pandemic. <https://tinyurl.com/cjkmbv2v>. 2021. [Press release]
5. Ratzki-Leewing A. Food Insecurity and Hypoglycemia. The Morning Show with Devon Peacock (D. Peacock, Host), 2023. [Radio interview]
6. Cam Buchan. Food insecurity doubles rate of severe hypoglycemia in adults with diabetes: Study. *Schulich School of Medicine & Dentistry, Western University*. <https://tinyurl.com/ytknr3b8>. 2023. [News article]
7. Food insecurity amplifies severe hypoglycemia risk in diabetic adults. *Open Access Government*. <https://tinyurl.com/2p8ucumw>. 2023. [News article].
8. Food insecurity doubles the rate of severe hypoglycaemia in adults with diabetes. *Diabetes Times*. <https://tinyurl.com/3fvh2b6s>. 2023. [News article].
9. Lisa Buckingham & EASD e-learning. Hypoglycemia in older adults with diabetes. *Horizons News*. <https://tinyurl.com/3typsf3z>. 2023. [News article]
10. Megan Craig. Study finds link between food insecurity and severe hypoglycemia. *News Medical*. <https://tinyurl.com/yc8aeyzh>. 2023. [News article]
11. Connection between food insecurity, rate of severe hypoglycaemia in adults with diabetes: Study. *The Times of India*. <https://tinyurl.com/2s4yuf58>. 2023. [News article]
12. EurekAlert! EASD Media Office. Food insecurity doubles rate of severe hypoglycaemia in adults with diabetes. <https://tinyurl.com/mwk435b4>. 2023. [News article]